

ARUNKUMAR K R

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🐙 GitHub | 💻 Leetcode | 🏠 Hackerrank

SUMMARY

Aspiring CSE student with hands-on internship experience in full-stack and machine learning engineering. Published researcher (YOLOv8/Computer Vision), RHCSA-certified, and active competitive programmer (500+ DSA problems). Builds production-ready systems using React, Node.js, FastAPI, and LLM pipelines. Seeking a software engineering role to deliver scalable, impact-driven solutions.

EDUCATION

KPR Institute of Engineering and Technology, Coimbatore	2023 – 2027
B.E. Computer Science and Engineering	CGPA: 8.77 / 10
Kongu Vellalar Matriculation Higher Secondary School, Perundurai	2022 – 2023
Higher Secondary Education (Class XII)	94.6%

TECHNICAL SKILLS

- **Languages & Core CS:** Python, Java, JavaScript (ES6+), C, SQL, DSA, OOP, DBMS
- **Web & DevOps:** React.js, Node.js, FastAPI, Flask, Streamlit, REST APIs, MongoDB, PostgreSQL, Docker, Git, GitHub, ChromaDB
- **AI / ML:** Deep Learning (CNNs, Transformers), RAG Pipelines, LangChain, TensorFlow, PyTorch, Scikit-Learn
- **Data & Analytics:** PowerBI, Data Modeling, Data Analytics, Pandas, Numpy

EXPERIENCE

Reccsar Pvt Ltd	Jan 2025 – Feb 2025
<i>Full Stack Developer Intern</i>	
• Developed and maintained full-stack web features using React.js on the frontend and Node.js/Express on the backend, contributing to live product features used by end users. • Integrated RESTful APIs and worked with relational databases to implement dynamic data-driven UI components, reducing manual data entry workflows.	

PROJECTS

ProManage – Collaborative Project Management API	Live Demo GitHub
<i>Node.js, TypeScript, PostgreSQL, Redis, JWT</i>	
• Architected a scalable REST API with Role-Based Access Control (RBAC) and JWT authentication, enforcing granular permission tiers across user roles (admin, manager, member). • Reduced API response latency by implementing Redis caching and Node.js Streams for efficient large file transfers, achieving consistent low-latency responses under simulated load.	
Prism AI – Content Generation Suite	Live Demo GitHub
<i>Python, FastAPI, Groq LLM, Hugging Face FLUX.1</i>	
• Built a multi-modal AI content platform generating SEO-optimized blog posts, video scripts, and social media images by integrating Groq LLM and Hugging Face FLUX models via async API calls. • Served a glassmorphism single-page frontend through a unified FastAPI backend, reducing infrastructure complexity while supporting multiple concurrent AI generation tasks.	
RouteX – Intelligent PDF Routing System	Live Demo GitHub
<i>Python, PyMuPDF, Tesseract OCR, NLP</i>	
• Built an intelligent document classification pipeline using hybrid text extraction (PyMuPDF + Tesseract OCR), correctly routing scanned and digital PDFs to departments with high accuracy. • Applied NLP-based summarization and TF-IDF keyword extraction for automatic document tagging, reducing manual categorization effort significantly.	

RESEARCH & ACHIEVEMENTS

- **Research Publication:** "Real-Time Surface Defect Detection in Metals using YOLOv8" – presented and published at the **3rd ICEAMST International Conference**; focused on industrial CV applications with deep learning.
- **RedHat Certified System Administrator (RHCSA)** – Industry-recognized Linux system certification.
- **Competitive Programming:** Solved **500+ DSA problems** on LeetCode and HackerRank, with verified certifications in Problem Solving (Intermediate), Python, Java, and SQL.
- **Leadership:** Executive Member at Dept. Association – coordinated technical events and workshops for 100+ students.